

Assessing Relative Agreement in Ranking Rhode Island Census Tracts Across the Social Vulnerability Index and the Enviornmental Justice Index

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Project Overview

This purpose of this document is to compare of national indices of place-based vulnerability within the state of Rhode Island. The two indices of interest include the Social Vulnerability Index (SVI) and the Environmental Justice Index (EJI), which are both created by the Centers for Disease Control and Prevention and the Agency for Toxic Substances and Disease Registry. Indexes are composite measures that produce singular value to rank census tracts with respect to varying definitions of vulnerability. Thus the goal for each index is to identify vulnerable communities in order to guide resources allocation and policy development to improve health and health equity. The SVI measures social vulnerability, which refers to socioeconomic factors that exacerbate the consequences communities face during natural/human-made disasters and infectious disease outbreaks. While EJI measures environmental justice refers to the capacity that all people, regardless of background have equitable protection from environmental hazards and access to environmental benefits. Therefore, SVI identifies communities who are at increased risk who are disproportionately affected during times of crisis due to socioeconomic factors, while the EJI measures the cumulative impact of environmental burden by incorporating environmental exposures.

The goal of this report is to quantify the relative agreement of these two different definitions of vulnerability. If the SVI and the EJI align, then these communities face both social and environmental burdens, thus resulting in heightened overall risk of adverse health outcomes. On the other hand if the SVI and EJI do not align then, this would suggest that specific communities are more socially vulnerable without being environmentally burdened, or vice-versa. The distinction between the two would determine the types of interventions required for communities across Rhode Island (i.e., prioritize interventions that mitigate both social and environmental burdens or tailored interventions).

- **Task 1:** Calculate Weighted Kappa to assess agreement between SVI and EJI across quantiles ranging from tertiles to deciles for Rhode Island census tracts.
- **Task 2:** Visualize the spatial distribution of index values across the state

Cleaning Indices and Feature Engineering Quantiles

Social Vulnerabilty Index Variables

Detailed description of 2022 SVI calculations and Documentations is available at [CDC/ATSDR SVI: Data and Documentation Download Website](#). Variables that will be pulled from the SVI include: the summary variable and the 4 theme specific variables (i.e., Socioeconomic Status, Household Characteristics, Racial & Ethnic Minority Status, Housing Type & Transportation). The figure below illustrates the American Community Survey, 2018-2022 (5-year) variables used for calculating the SVI.

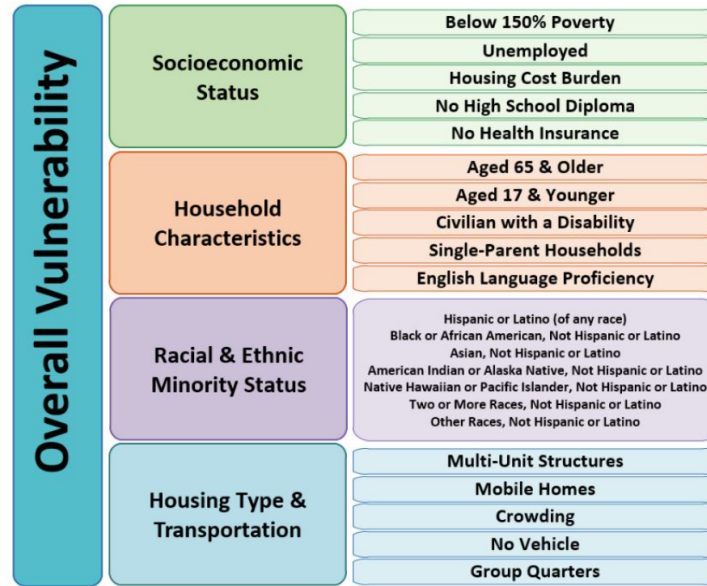


Figure 1: Organizational Structure of SVI

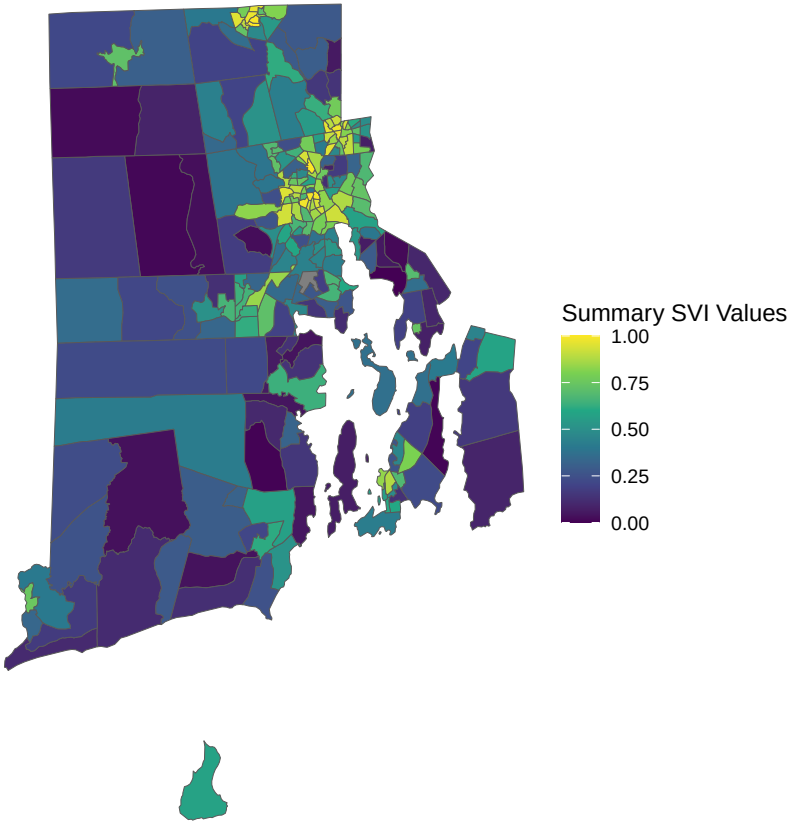
Overview of Methods:

- Data Cleaning:
 - Selected relevant columns: FIPS code, location, and SVI theme variables (prefix rpl_).
 - Filtered out rows with invalid values (-999) in the SVI themes to ensure data integrity.
 - * Cells with values of -999 were dropped because this means that the value was unavailable from the original census data or we could not calculate a derived value because of unavailable Census data.
- Exploratory Data Analysis (EDA):
 - Generated summary statistics for SVI theme variables to provide an overview of data distribution, indicating relative consistency among the values.
 - Visualize the spatial distribution of the variables # Quantile Calculation:
 - Developed a function to compute quantiles for the rpl_themes variable, calculating thirds, fourths, fifths, sixths, sevenths, eighths, ninths, and tenths.
 - Applied the quantile calculations to categorize SVI values into distinct groups based on their respective quantile thresholds.
 - Converted the resulting categorical variables into factors for subsequent analysis.

Table 1: 5 Point Summary of SVI Index Values

Statistic	Value
Minimum	0.00
1st Quartile	0.25
Median	0.50
Mean	0.50
3rd Quartile	0.75
Maximum	1.00

Map 1: Spatial Distribution of Summary SVI Values, 2022



Enviornmental Justice Index

Detailed description of 2022 EJI calculations and Documentations is available at [CDC/ATSDR SVI:EJI Technical Documentation Website](#). The variables extracted from the EJI include the overall summary value and three EJI theme-specific variables: Environmental Burden, Social Vulnerability, and Health Vulnerability. Unlike the SVI, which calculates index values within each state, the state-level EJI values are calculated using the United States as a whole as the reference group. To illustrate this concept, consider the following quote from the EJI Data Download page: “For example, an EJI ranking of 0.85 signifies that 85% of tracts in the nation likely experience less severe cumulative impacts from environmental burden than the tract of interest, and that 15% of tracts in the nation likely experience more severe cumulative impacts from environmental burden.” To ensure consistency when comparing the SVI and EJI, the EJI values needed to be recalculated so that the index values are aligned.

Social Vulnerability	Racial/ Ethnic Minority Status	Minority Status
	Socioeconomic Status	Poverty
		No High School Diploma
		Unemployment
		Housing Tenure
		Housing Burdened Lower-Income Households
		Lack of Health Insurance
		Lack of Broadband Access
	Household Characteristics	Age 65 and Older
		Age 17 and Younger
		Civilian with a Disability
		Speaks English “Less than Well”
Environmental Burden	Housing Type	Group Quarters
		Mobile Homes
	Air Pollution	Ozone
		PM2.5
		Diesel Particulate Matter
		Air Toxics Cancer Risk
	Potentially Hazardous & Toxic Sites	National Priority List Sites
		Toxic Release Inventory Sites
		Treatment, Storage, and Disposal Sites
		Risk Management Plan Sites
		Coal Mines
	Built Environment	Lead Mines
		Lack of Recreational Parks
		Houses Built Pre-1980
	Transportation Infrastructure	Lack of Walkability
		High-Volume Roads
		Railways
Health Vulnerability	Pre-existing Chronic Disease Burden	Airports
		Water Pollution
		Impaired Surface Water
		Asthma*
		Cancer*
		High Blood Pressure*
		Diabetes*
		Poor Mental Health*

Figure 2: Organizational Structure of EJI

Overview of Methods:

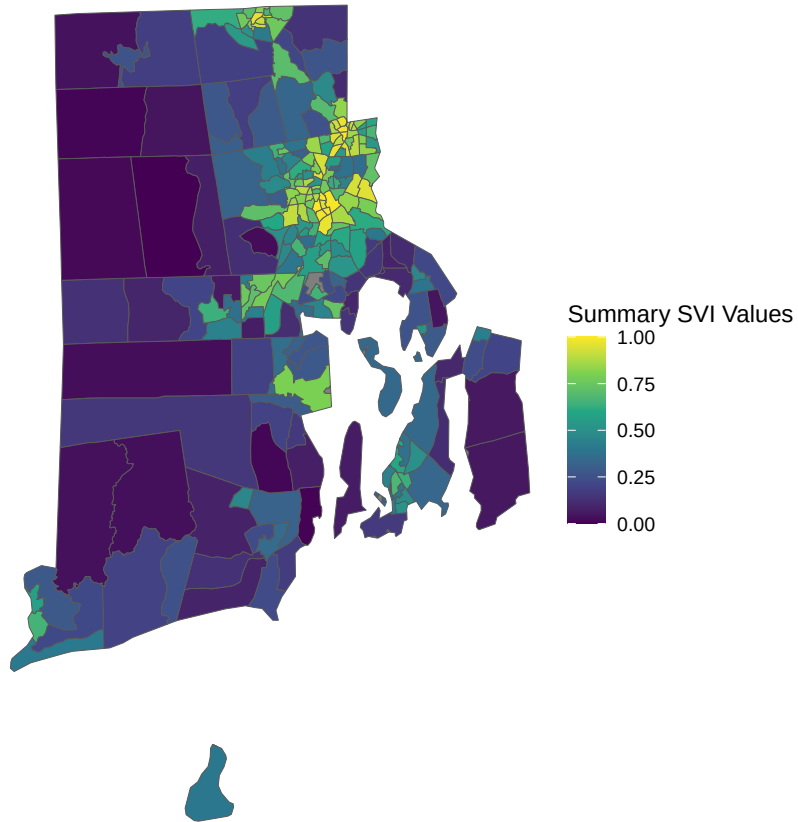
- Module 1 & 2: Environmental Burden - calculate percentile ranks
- Module 3: Health Vulnerability - calculate percentile ranks

- Created threshold variables for asthma, cancer, high blood pressure, diabetes, and mental health, based on quantiles of each health variable.
- Summed the threshold flags to obtain a flag_score and computed the health vulnerability score (rpl_mod3) by multiplying the flag_score by 0.20.
- Combining Modules:
 - Merged the results from Module 1, Module 2, and Module 3 using left joins on geoid.
 - Calculated a total rank score by summing the numeric columns across the merged dataset by calculating percentile ranks
- Quantile Generation - Generated quantiles for the rpl_eji scores and created categorical variables (thirds, fourths, fifths, etc.) based on the computed quantiles.

Table 2: 5 Point Summary of EJI Index Values

Statistic	Value
Minimum	0.00
1st Quartile	0.25
Median	0.50
Mean	0.50
3rd Quartile	0.75
Maximum	1.00

Map 2: Spatial Distribution of Summary EJI Values, 2022



Identification of the Missing Census Tracts

The Economic Justice Index (EJI) uses 2010 FIPS codes, whereas the Social Vulnerability Index (SVI) employs 2022 FIPS codes, resulting in misalignment across spatial definitions. While the larger EJI tracts encompass two smaller current SVI tracts, the smaller SVI summary values cannot be combined due to their representation as percentile ranks, making attempts to calculate weighted Cohen's kappa potentially incorrect or inaccurate. One cannot simply sum the individual estimates used to calculate the Economic Justice Index (EJI) because the index is based on modeled estimates that may have different underlying distributions and variances, potentially introducing additional bias. Summing these values without accounting for their unique modeling contexts could distort the final index values and lead to inaccurate representations of the underlying socioeconomic conditions.

```
## [1] "12 Census Tracts that appear in the SVI but do not appear in the EJI"
```

```
## # A tibble: 12 x 2
```

```
##   fips      location
```

```
##   <chr>      <chr>
```

```
## 1 44005040104 Census Tract 401.04; Newport County; Rhode Island
```

```
## 2 44005040105 Census Tract 401.05; Newport County; Rhode Island
```

```
## 3 44007000301 Census Tract 3.01; Providence County; Rhode Island
```

```
## 4 44007000302 Census Tract 3.02; Providence County; Rhode Island
```

```
## 5 44007001601 Census Tract 16.01; Providence County; Rhode Island
```

```
## 6 44007001602 Census Tract 16.02; Providence County; Rhode Island
```

```
## 7 44007002801 Census Tract 28.01; Providence County; Rhode Island
```

```
## 8 44007002802 Census Tract 28.02; Providence County; Rhode Island
```

```
## 9 44007011404 Census Tract 114.04; Providence County; Rhode Island
```

```
## 10 44007011405 Census Tract 114.05; Providence County; Rhode Island
```

```
## 11 44009050701 Census Tract 507.01; Washington County; Rhode Island
```

```
## 12 44009050702 Census Tract 507.02; Washington County; Rhode Island
```

```
## [1] "6 Census Tracts that appear in the EJI but do not appear in the SVI"
```

```
## # A tibble: 6 x 2
```

```
##   geoid      location
```

```
##   <chr>      <chr>
```

```
## 1 44005040103 Census Tract 401.03, Newport County, Rhode Islan
```

```
## 2 44007000300 Census Tract 3, Providence County, Rhode Island
```

```
## 3 44007001600 Census Tract 16, Providence County, Rhode Island
```

```
## 4 44007002800 Census Tract 28, Providence County, Rhode Island
```

```
## 5 44007011401 Census Tract 114.01, Providence County, Rhode Is
```

```
## 6 44009050700 Census Tract 507, Washington County, Rhode Islan
```


Calculating Kappa Coefficient for Overall Indices Value

Weighted Cohen's kappa is a measure of the agreement between two ordinarily scaled samples, this is the variable estimate of interest because the samples are ordinal. Assumptions for Weighted Cohen's Kappa include:

1. You have two outcome categorical variables, which should be ordinal
2. The two outcome variables should have exactly the same categories
3. You have paired observations; each subject is categorized twice by two independent raters or methods.
4. The same two raters are used for all participants.

Statistical Hypothesis include:

- **Null Hypothesis:** $\kappa = 0$
- **Alternative Hypothesis:** $\kappa \neq 0$

The following interpretations will be implemented:

- values greater than 0.75 or so may be taken to represent excellent agreement beyond chance,
- values below 0.40 or so may be taken to represent poor agreement beyond chance, and
- values between 0.40 and 0.75 may be taken to represent fair to good agreement beyond chance.

Joseph L. Fleiss, Myunghee Cho Paik, Bruce Levin. 2003. Statistical Methods for Rates and Proportions. 3rd ed. John Wiley; Sons, Inc.

Quantiles	Weighted κ (95% CI)
Terciles	0.78 (0.73, 0.83)
Quartiles	0.78 (0.73, 0.83)
Quintiles	0.77 (0.71, 0.82)
Sextiles	0.79 (0.74, 0.84)
Septiles	0.8 (0.74, 0.85)
Octiles	0.82 (0.77, 0.86)
Nineths	0.8 (0.76, 0.85)
Deciles	0.8 (0.75, 0.85)

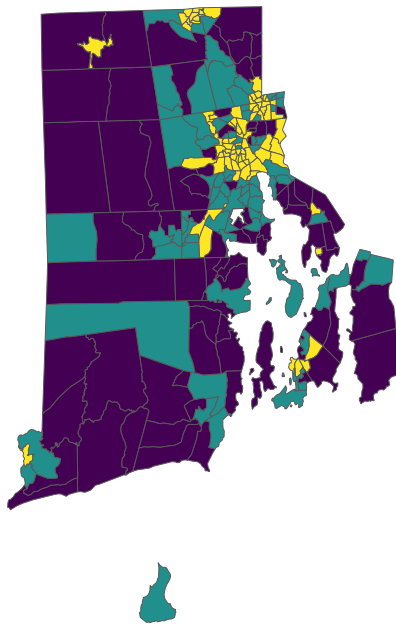
There is excellent agreement across all indices regardless of the quantiles being used, index agreement doe not change as when increasing the number of quantiles.

Appendix A - Spatial Distribution of Index Values

SVI Maps Across Tercile and Decile

Map 3: Quantile Maps of Social Vulnerability Index, 2022

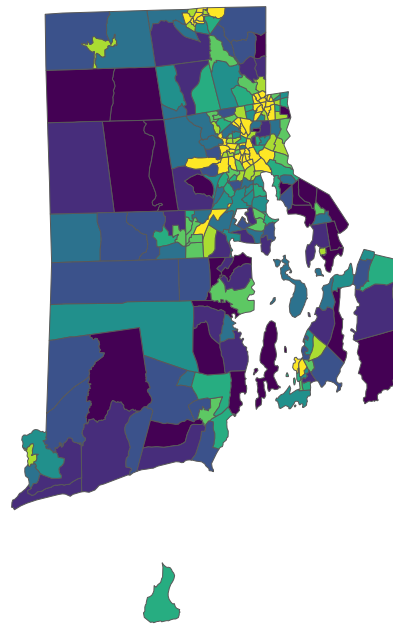
Tertiles



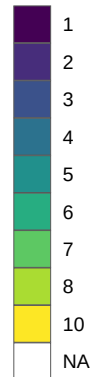
Summary Values



Deciles



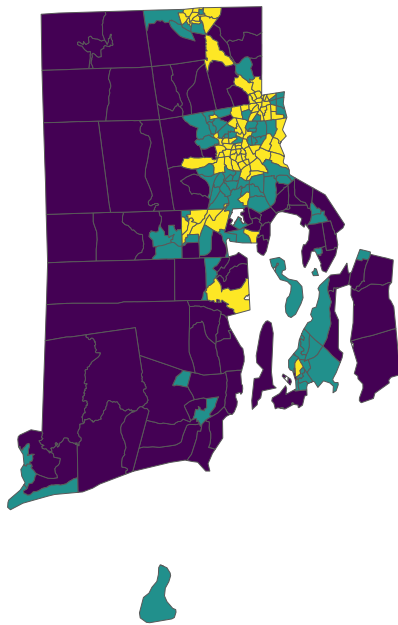
Summary Values



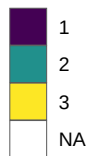
EJI Maps Across Tercile and Decile

Map 4: Quantile Maps of Enviornmental Justice Index, 2022

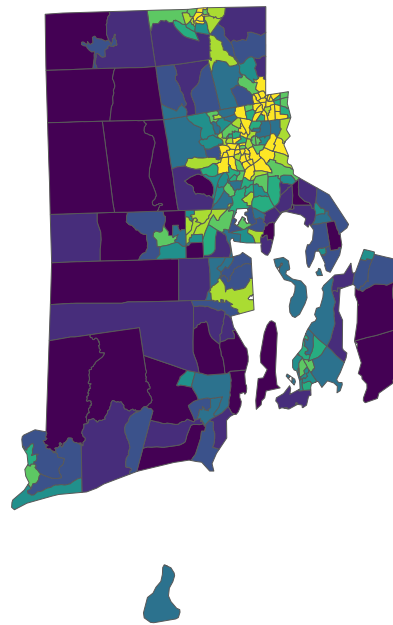
Tertiles



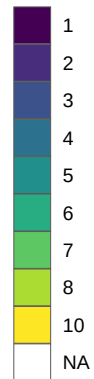
Summary Values



Deciles



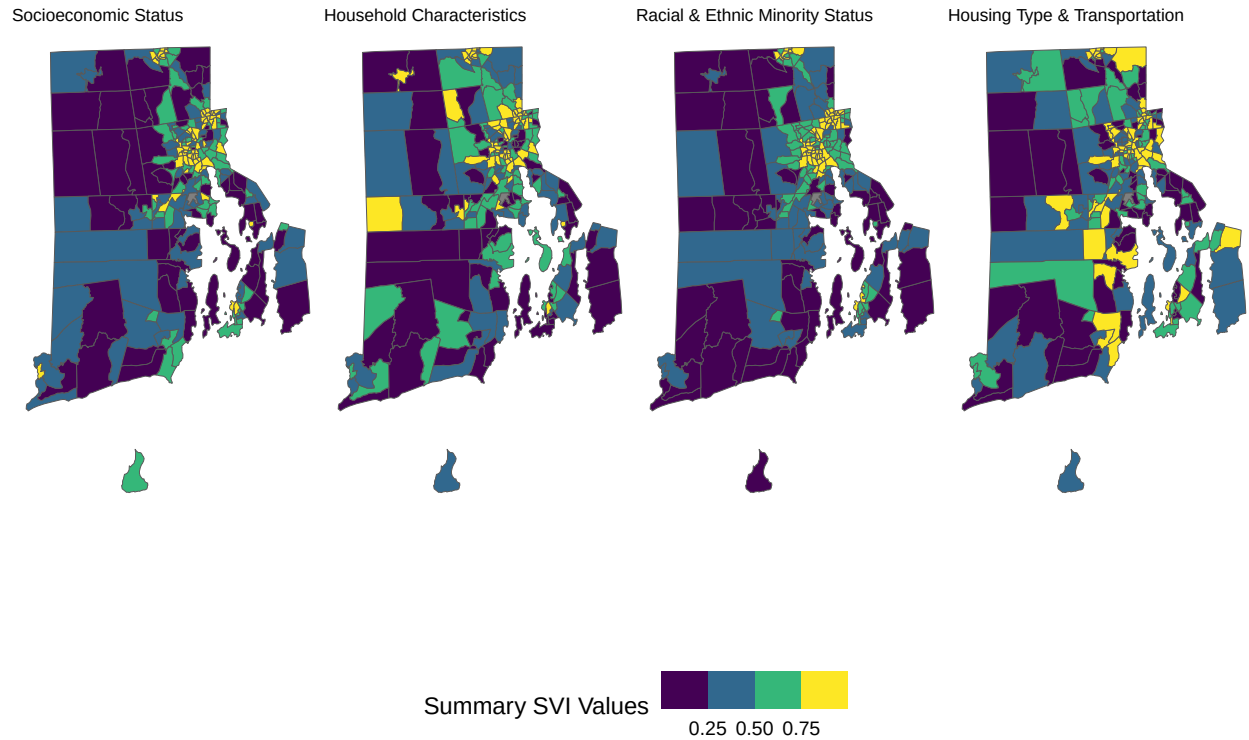
Summary Values



Distribution of Themes within Each Indices

The SVI contains 4 themes, and the distribution of those themes are shown below. Quantiles of these values will be made available upon request.

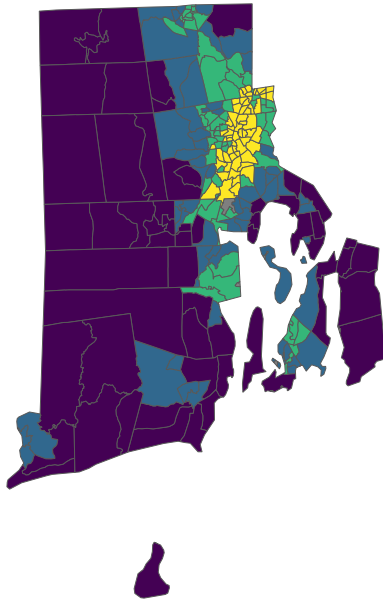
Map 5: Maps of Social Vulnerability Index Themes, 2022



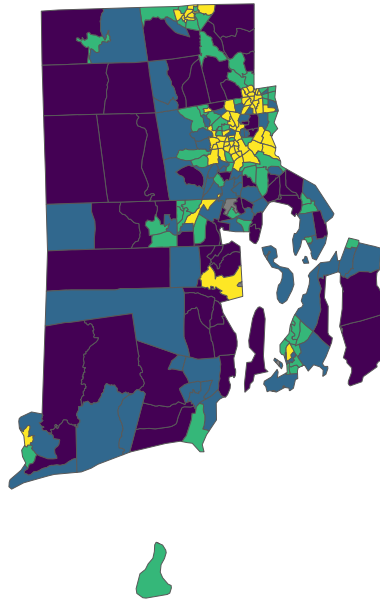
The EJI contains 3 themes, and the distribution of those themes are shown below. Quantiles of these values will be made available upon request.

Map 6: Maps of Envirovnmental Justice Index Themes, 2022

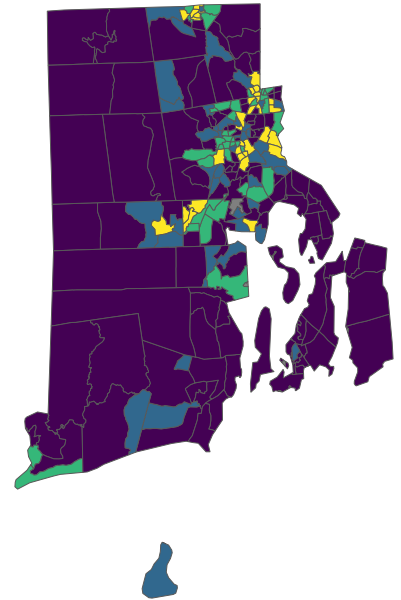
Envirovnmental Burden



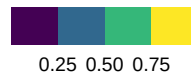
Social Vulnerability



Health Vulnerability



Summary SVI Values



Code Appendix

```
### importing packages
options(repos = c(CRAN = "https://cran.r-project.org"))
knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE,
  fig.width = 8,
  fig.height = 6)

# define desired packages
library(tidyverse) # general data manipulation
library(here)      # define top level of project folder
                  # this allows for specification of where
                  # things live in relation to the top level

library(gtools)
library(psych)
library(ggpubr)
library(gridExtra)
library(lemon)
library(janitor)
library(kableExtra)
# spatial tasks
library(tigris)    # obtain shp files

### loading data
svi = read_csv(here::here("01_Data", "SVI_RI_Data.csv")) %>% clean_names() %>%
  mutate(fips = as.character(fips))
eji = read_csv(here::here("01_Data", "EJI_RI_Data.csv")) %>% clean_names() %>%
  mutate(geoid = as.character(geoid))

## defining themes
envi_burden = c("e_ozone", "e_pm", "e_dslpm", "e_totcr", "e_npl", "e_tri", "e_tsd", "e_rmp", "e_
social_burden = c("ep_minrty", "ep_pov200", "ep_nohsdp", "ep_unemp", "ep_renter", "ep_houbdn", "
health_vul = c("ep_asthma", "ep_cancer", "ep_bphigh", "ep_diabetes", "ep_mhlth")

# obtaining SPH files for RI tracts
ri_tracts = tracts(state = "RI", year = 2022, cb = TRUE)
#### Cleaning
df_svi = svi %>% select(fips, location, starts_with("rpl_"))
# dropping variables that have -999 as the value
df_svi = df_svi %>%
  filter_at(vars(rpl_themes, rpl_theme1, rpl_theme2,
    rpl_theme3, rpl_theme4),
    any_vars(. != -999))

#### Brief EDA - no need for graphics since the table illustrates
# the values are relatively the same
df_svi %>% select(starts_with("rpl_")) %>% summary()
```

Creating Quantiles

```
generate_quantiles = function(data, column_name) {  
  # Compute quantiles for different levels  
  quantiles = list(  
    thirds = quantile(data[[column_name]], probs = c(0, 1/3, 2/3, 1), na.rm = TRUE),  
    fourths = quantile(data[[column_name]], probs = c(0, 1/4, 2/4, 3/4, 1), na.rm = TRUE),  
    fifths = quantile(data[[column_name]], probs = c(0, 1/5, 2/5, 3/5, 4/5, 1), na.rm = TRUE),  
    sixths = quantile(data[[column_name]], probs = c(0, 1/6, 2/6, 3/6, 4/6, 5/6, 1), na.rm = TRUE),  
    sevenths = quantile(data[[column_name]], probs = c(0, 1/7, 2/7, 3/7, 4/7, 5/7, 6/7, 1), na.rm = TRUE),  
    eighths = quantile(data[[column_name]], probs = c(0, 1/8, 2/8, 3/8, 4/8, 5/8, 6/8, 7/8, 1), na.rm = TRUE),  
    ninths = quantile(data[[column_name]], probs = c(0, 1/9, 2/9, 3/9, 4/9, 5/9, 6/9, 7/9, 8/9, 1), na.rm = TRUE),  
    tenths = quantile(data[[column_name]], probs = c(0, 1/10, 2/10, 3/10, 4/10, 5/10, 6/10, 7/10, 8/10, 9/10, 1), na.rm = TRUE)  
  )  
  
  # Return the list of quantiles  
  return(quantiles)  
}
```

```
# Apply the quantile function to each column and bind the results  
svi_quantiles = generate_quantiles(df_svi, "rpl_themes")
```

```
svi_quantiles = df_svi %>%  
  select(fips, location, rpl_themes) %>%  
  mutate(  
    thirds = case_when(  
      rpl_themes <= svi_quantiles$thirds[2] ~ 1, # Below or equal to 33.33%  
      rpl_themes <= svi_quantiles$thirds[3] ~ 2, # Between 33.33% and 66.67%  
      rpl_themes <= svi_quantiles$thirds[4] ~ 3, # Above 66.67%  
    ),  
    fourths = case_when(  
      rpl_themes <= svi_quantiles$fourths[2] ~ 1,  
      rpl_themes <= svi_quantiles$fourths[3] ~ 2,  
      rpl_themes <= svi_quantiles$fourths[4] ~ 3,  
      rpl_themes <= svi_quantiles$fourths[5] ~ 4,  
    ),  
    fifths = case_when(  
      rpl_themes <= svi_quantiles$fifths[2] ~ 1,  
      rpl_themes <= svi_quantiles$fifths[3] ~ 2,  
      rpl_themes <= svi_quantiles$fifths[4] ~ 3,  
      rpl_themes <= svi_quantiles$fifths[5] ~ 4,  
      rpl_themes <= svi_quantiles$fifths[6] ~ 5,  
    ),  
    sixths = case_when(  
      rpl_themes <= svi_quantiles$sixths[2] ~ 1,  
      rpl_themes <= svi_quantiles$sixths[3] ~ 2,  
      rpl_themes <= svi_quantiles$sixths[4] ~ 3,  
      rpl_themes <= svi_quantiles$sixths[5] ~ 4,  
      rpl_themes <= svi_quantiles$sixths[6] ~ 5,  
    )  
  )
```

```

    rpl_themes <= svi_quantiles$sixths[7] ~ 6),
sevenths = case_when(
  rpl_themes <= svi_quantiles$sevenths[2] ~ 1,
  rpl_themes <= svi_quantiles$sevenths[3] ~ 2,
  rpl_themes <= svi_quantiles$sevenths[4] ~ 3,
  rpl_themes <= svi_quantiles$sevenths[5] ~ 4,
  rpl_themes <= svi_quantiles$sevenths[6] ~ 5,
  rpl_themes <= svi_quantiles$sevenths[7] ~ 6,
  rpl_themes <= svi_quantiles$sevenths[8] ~ 7),
eighths = case_when(
  rpl_themes <= svi_quantiles$eighths[2] ~ 1,
  rpl_themes <= svi_quantiles$eighths[3] ~ 2,
  rpl_themes <= svi_quantiles$eighths[4] ~ 3,
  rpl_themes <= svi_quantiles$eighths[5] ~ 4,
  rpl_themes <= svi_quantiles$eighths[6] ~ 5,
  rpl_themes <= svi_quantiles$eighths[7] ~ 6,
  rpl_themes <= svi_quantiles$eighths[8] ~ 7,
  rpl_themes <= svi_quantiles$eighths[9] ~ 8),
ninths = case_when(
  rpl_themes <= svi_quantiles$ninths[2] ~ 1,
  rpl_themes <= svi_quantiles$ninths[3] ~ 2,
  rpl_themes <= svi_quantiles$ninths[4] ~ 3,
  rpl_themes <= svi_quantiles$ninths[5] ~ 4,
  rpl_themes <= svi_quantiles$ninths[6] ~ 5,
  rpl_themes <= svi_quantiles$ninths[7] ~ 6,
  rpl_themes <= svi_quantiles$ninths[8] ~ 7,
  rpl_themes <= svi_quantiles$ninths[9] ~ 8,
  rpl_themes <= svi_quantiles$ninths[10] ~ 9),
tenths = case_when(
  rpl_themes <= svi_quantiles$tenths[2] ~ 1,
  rpl_themes <= svi_quantiles$tenths[3] ~ 2,
  rpl_themes <= svi_quantiles$tenths[4] ~ 3,
  rpl_themes <= svi_quantiles$tenths[5] ~ 4,
  rpl_themes <= svi_quantiles$tenths[6] ~ 5,
  rpl_themes <= svi_quantiles$tenths[7] ~ 6,
  rpl_themes <= svi_quantiles$tenths[8] ~ 7,
  rpl_themes <= svi_quantiles$tenths[9] ~ 8,
  rpl_themes <= svi_quantiles$tenths[11] ~ 10)
)

names = svi_quantiles %>% select(-fips, -rpl_themes, -location) %>% colnames()

svi_quantiles = svi_quantiles %>%
  mutate(across(any_of(names), ~as.factor(.)))
svi_quantiles %>%
  reframe(
    Minimum = min(rpl_themes, na.rm = TRUE),

```



```

    "1st Quartile" = quantile(rpl_themes, 0.25, na.rm = TRUE),
    Median = median(rpl_themes, na.rm = TRUE),
    Mean = mean(rpl_themes, na.rm = TRUE),
    "3rd Quartile" = quantile(rpl_themes, 0.75, na.rm = TRUE),
    Maximum = max(rpl_themes, na.rm = TRUE)
  ) %>%
  pivot_longer(cols = everything(), names_to = "Statistic", values_to = "Value") %>%
  kable(digits = 2, caption = "5 Point Summary of SVI Index Values")
# joining data
svi_map = left_join(ri_tracts, svi_quantiles, join_by(GEOID == fips))

# making maps
ggplot(data = svi_map, aes(fill = rpl_themes)) +
  geom_sf(lwd = 0.01) +
  theme_void() +
  labs(title = "Map 1: Spatial Distribution of Summary SVI Values, 2022", fill = "Summary SVI")
  scale_fill_viridis_c()
module_1 = eji %>%
  select(geoid, location, any_of(envi_burden)) %>%
  mutate(across(where(is.numeric), ~percent_rank(.), .names = "percentile_{.col}")) %>%
  select(geoid, location, starts_with("percentile_")) %>%
  rowwise() %>%
  mutate(percentile_rank_sum = sum(c_across(where(is.numeric)))) %>%
  ungroup() %>%
  mutate(rpl_mod1 = round(percent_rank(percentile_rank_sum), 4)) %>%
  select(geoid, location, rpl_mod1)

module_2 = eji %>%
  select(geoid, location, any_of(social_burden)) %>%
  mutate(across(where(is.numeric), ~percent_rank(.), .names = "percentile_{.col}")) %>%
  select(geoid, location, starts_with("percentile_")) %>%
  rowwise() %>%
  mutate(percentile_rank_sum = sum(c_across(where(is.numeric)))) %>%
  ungroup() %>%
  mutate(rpl_mod2 = round(percent_rank(percentile_rank_sum), 4)) %>%
  select(geoid, location, rpl_mod2)

# Example usage with eji dataset and specified health vulnerability columns
module_3 = eji %>%
  select(geoid, location, any_of(health_vul)) %>%
  mutate(threshold_asthma = cut(ep_asthma,
                                breaks = quantile(ep_asthma, probs = c(0, 1/3, 2/3, 1), na.rm = TRUE),
                                include.lowest = TRUE,
                                labels = c(0, 0, 1)),
         threshold_cancer = cut(ep_cancer,
                                breaks = quantile(ep_cancer, probs = c(0, 1/3, 2/3, 1), na.rm = TRUE),
                                include.lowest = TRUE,

```

```

        labels = c(0, 0, 1)),
threshold_bphigh = cut(ep_bphigh,
        breaks = quantile(ep_bphigh, probs = c(0, 1/3, 2/3, 1), na.rm =
        include.lowest = TRUE,
        labels = c(0, 0, 1)),
threshold_diabetes = cut(ep_diabetes,
        breaks = quantile(ep_diabetes, probs = c(0, 1/3, 2/3, 1), na.rm =
        include.lowest = TRUE,
        labels = c(0, 0, 1)),
threshold_mhlth = cut(ep_mhlth,
        breaks = quantile(ep_mhlth, probs = c(0, 1/3, 2/3, 1), na.rm =
        include.lowest = TRUE,
        labels = c(0, 0, 1))) %>%
select(geoid, location, starts_with("threshold_")) %>%
mutate(across(starts_with("threshold_"), ~ as.numeric(as.character(.)))) %>%
rowwise() %>%
mutate(flag_score = sum(c_across(where(is.numeric)))) %>%
ungroup() %>%
mutate(rpl_mod3 = round(flag_score * 0.20, 2)) %>%
select(geoid, location, rpl_mod3)

df_eji = module_1 %>%
left_join(module_2, by = "geoid") %>%
left_join(module_3, by = "geoid") %>%
rowwise() %>%
mutate(eji_sum_rank = sum(c_across(where(is.numeric)))) %>%
ungroup() %>%
mutate(rpl_eji = round(percent_rank(eji_sum_rank),4)) %>%
select(-eji_sum_rank)

# Apply the quantile function to each column and bind the results
eji_quantiles = generate_quantiles(df_eji, "rpl_eji")

eji_quantiles = df_eji %>%
select(geoid, location, rpl_eji) %>%
mutate(
  thirds = case_when(
    rpl_eji <= eji_quantiles$thirds[2] ~ 1, # Below or equal to 33.33%
    rpl_eji <= eji_quantiles$thirds[3] ~ 2, # Between 33.33% and 66.67%
    rpl_eji <= eji_quantiles$thirds[4] ~ 3 # Above 66.67%
  ),
  fourths = case_when(
    rpl_eji <= eji_quantiles$fourths[2] ~ 1,
    rpl_eji <= eji_quantiles$fourths[3] ~ 2,
    rpl_eji <= eji_quantiles$fourths[4] ~ 3,
    rpl_eji <= eji_quantiles$fourths[5] ~ 4),

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fifths = case_when(
  rpl_eji <= eji_quantiles$fifths[2] ~ 1,
  rpl_eji <= eji_quantiles$fifths[3] ~ 2,
  rpl_eji <= eji_quantiles$fifths[4] ~ 3,
  rpl_eji <= eji_quantiles$fifths[5] ~ 4,
  rpl_eji <= eji_quantiles$fifths[6] ~ 5),
sixths = case_when(
  rpl_eji <= eji_quantiles$sixths[2] ~ 1,
  rpl_eji <= eji_quantiles$sixths[3] ~ 2,
  rpl_eji <= eji_quantiles$sixths[4] ~ 3,
  rpl_eji <= eji_quantiles$sixths[5] ~ 4,
  rpl_eji <= eji_quantiles$sixths[6] ~ 5,
  rpl_eji <= eji_quantiles$sixths[7] ~ 6),
sevenths = case_when(
  rpl_eji <= eji_quantiles$sevenths[2] ~ 1,
  rpl_eji <= eji_quantiles$sevenths[3] ~ 2,
  rpl_eji <= eji_quantiles$sevenths[4] ~ 3,
  rpl_eji <= eji_quantiles$sevenths[5] ~ 4,
  rpl_eji <= eji_quantiles$sevenths[6] ~ 5,
  rpl_eji <= eji_quantiles$sevenths[7] ~ 6,
  rpl_eji <= eji_quantiles$sevenths[8] ~ 7),
eighths = case_when(
  rpl_eji <= eji_quantiles$eighths[2] ~ 1,
  rpl_eji <= eji_quantiles$eighths[3] ~ 2,
  rpl_eji <= eji_quantiles$eighths[4] ~ 3,
  rpl_eji <= eji_quantiles$eighths[5] ~ 4,
  rpl_eji <= eji_quantiles$eighths[6] ~ 5,
  rpl_eji <= eji_quantiles$eighths[7] ~ 6,
  rpl_eji <= eji_quantiles$eighths[8] ~ 7,
  rpl_eji <= eji_quantiles$eighths[9] ~ 8),
ninths = case_when(
  rpl_eji <= eji_quantiles$ninths[2] ~ 1,
  rpl_eji <= eji_quantiles$ninths[3] ~ 2,
  rpl_eji <= eji_quantiles$ninths[4] ~ 3,
  rpl_eji <= eji_quantiles$ninths[5] ~ 4,
  rpl_eji <= eji_quantiles$ninths[6] ~ 5,
  rpl_eji <= eji_quantiles$ninths[7] ~ 6,
  rpl_eji <= eji_quantiles$ninths[8] ~ 7,
  rpl_eji <= eji_quantiles$ninths[9] ~ 8,
  rpl_eji <= eji_quantiles$ninths[10] ~ 9),
tenths = case_when(
  rpl_eji <= eji_quantiles$tenths[2] ~ 1,
  rpl_eji <= eji_quantiles$tenths[3] ~ 2,
  rpl_eji <= eji_quantiles$tenths[4] ~ 3,
  rpl_eji <= eji_quantiles$tenths[5] ~ 4,
  rpl_eji <= eji_quantiles$tenths[6] ~ 5,
  rpl_eji <= eji_quantiles$tenths[7] ~ 6,

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    rpl_eji <= eji_quantiles$tenths[8] ~ 7,
    rpl_eji <= eji_quantiles$tenths[9] ~ 8,
    rpl_eji <= eji_quantiles$tenths[11] ~ 10)
  )

names = eji_quantiles %>% select(-geoid, -rpl_eji, -location) %>% colnames()

ej_i_quantiles = eji_quantiles %>%
  mutate(across(any_of(names), ~as.factor(.)))

ej_i_quantiles = eji_quantiles[complete.cases(eji_quantiles), ]
ej_i_quantiles %>%
  reframe(
    Minimum = min(rpl_eji, na.rm = TRUE),
    "1st Quartile" = quantile(rpl_eji, 0.25, na.rm = TRUE),
    Median = median(rpl_eji, na.rm = TRUE),
    Mean = mean(rpl_eji, na.rm = TRUE),
    "3rd Quartile" = quantile(rpl_eji, 0.75, na.rm = TRUE),
    Maximum = max(rpl_eji, na.rm = TRUE)
  ) %>%
  pivot_longer(cols = everything(), names_to = "Statistic", values_to = "Value") %>%
  kable(digits = 2, caption = "5 Point Summary of EJI Index Values")
# obtaining SPH files for RI tracts
ri_tracts = tracts(state = "RI", year = 2010, cb = TRUE)

# removing excess characters to allow for join
ri_tracts$GEO_ID = str_remove(ri_tracts$GEO_ID, "^1400000US")
# joining data
ej_i_map = left_join(x = ri_tracts, y = ej_i_quantiles, join_by(GEO_ID == geoid))

# making maps
ggplot(data = ej_i_map, aes(fill = rpl_eji)) +
  geom_sf(lwd = 0.01) +
  theme_void() +
  labs(title = "Map 2: Spatial Distribution of Summary EJI Values, 2022", fill = "Summary SVI")
  scale_fill_viridis_c()
# 12 census tracts appear in the svi but do not appear in the eji
print("12 Census Tracts that appear in the SVI but do not appear in the EJI")
anti_join(x = svi_quantiles, y = ej_i_quantiles, by = join_by(fips == geoid)) %>% select(fips, )

# 6 census tracts appear in the eji but do not appear in the svi
print("6 Census Tracts that appear in the EJI but do not appear in the SVI")
anti_join(x = ej_i_quantiles, y = svi_quantiles, by = join_by(geoid == fips)) %>% select(geoid, )
df = inner_join(svi_quantiles, y = ej_i_quantiles, by = join_by(fips == geoid))

a = cohen.kappa(cbind(df$fourths.x, df$fourths.y))
b = cohen.kappa(cbind(df$fourths.x, df$fourths.y))

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c = cohen.kappa(cbind(df$fifths.x, df$fifths.y))
d = cohen.kappa(cbind(df$sixths.x, df$sixths.y))
e = cohen.kappa(cbind(df$sevenths.x, df$sevenths.y))
f = cohen.kappa(cbind(df$eighths.x, df$eighths.y))
g = cohen.kappa(cbind(df$ninths.x, df$ninths.y))
h = cohen.kappa(cbind(df$tenths.x, df$tenths.y))
a = ggplot(data = svi_map, aes(fill = thirds)) +
  geom_sf(lwd = 0.1) +
  theme_void() +
  labs(title = "Tertiles",
       fill = "Summary Values") +
  scale_fill_viridis_d()

b = ggplot(data = svi_map, aes(fill = tenths)) +
  geom_sf(lwd = 0.1) +
  theme_void() +
  labs(title = "Deciles",
       fill = "Summary Values") +
  scale_fill_viridis_d()

plt_svi = ggarrange(a,b, ncol = 2)
annotate_figure(plt_svi, top = text_grob("Map 3: Quantile Maps of Social Vulnerability Index, ",
                                         face = "bold", size = 14))
a = ggplot(data = eji_map, aes(fill = thirds)) +
  geom_sf(lwd = 0.1) +
  theme_void() +
  labs(title = "Tertiles",
       fill = "Summary Values") +
  scale_fill_viridis_d()

b = ggplot(data = eji_map, aes(fill = tenths)) +
  geom_sf(lwd = 0.1) +
  theme_void() +
  labs(title = "Deciles",
       fill = "Summary Values") +
  scale_fill_viridis_d()

plt_eji = ggarrange(a,b, ncol = 2)
annotate_figure(plt_eji, top = text_grob("Map 4: Quantile Maps of Enviornmental Justice Index, ",
                                         face = "bold", size = 14))
# join theme values to svi_map
df_svi = df_svi %>% select(fips, rpl_theme1:rpl_theme4)
svi_map = left_join(svi_map, df_svi, by = join_by(GEOID == fips))
# make maps
a = ggplot(data = svi_map, aes(fill = rpl_theme1)) +
  geom_sf(lwd = 0.01) +
  theme_void() +
  theme(plot.title = element_text(size = 8)) +

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labs(title = "Socioeconomic Status", fill = "Summary SVI Values") +
scale_fill_viridis_b()

b = ggplot(data = svi_map, aes(fill = rpl_theme2)) +
  geom_sf(lwd = 0.01) +
  theme_void() +
  theme(plot.title = element_text(size = 8)) +
  labs(title = "Household Characteristics", fill = "Summary SVI Values") +
  scale_fill_viridis_b()

c = ggplot(data = svi_map, aes(fill = rpl_theme3)) +
  geom_sf(lwd = 0.01) +
  theme_void() +
  theme(plot.title = element_text(size = 8)) +
  labs(title = "Racial & Ethnic Minority Status", fill = "Summary SVI Values") +
  scale_fill_viridis_b()

d = ggplot(data = svi_map, aes(fill = rpl_theme4)) +
  geom_sf(lwd = 0.01) +
  theme_void() +
  theme(plot.title = element_text(size = 8)) +
  labs(title = "Housing Type & Transportation", fill = "Summary SVI Values") +
  scale_fill_viridis_b()

# join maps on one shared document
legend = lemon::g_legend(a + theme(legend.position='bottom'))
plt_svi_themes = ggarrange(
  a + theme(legend.position = "none"),
  b + theme(legend.position = "none"),
  c + theme(legend.position = "none"),
  d + theme(legend.position = "none"),
  ncol = 4)
annotate_figure(plt_svi_themes, top = text_grob("Map 5: Maps of Social Vulnerability Index Themes"))

# join themes to eji_map
df_eji = df_eji %>% select(geoid, starts_with("rpl_"), -rpl_eji)
eji_map = left_join(eji_map, df_eji, by = join_by(GEO_ID == geoid))

a = ggplot(data = eji_map, aes(fill = rpl_mod1)) +
  geom_sf(lwd = 0.1) +
  theme_void() +
  labs(title = "Environmental Burden", fill = "Environmental Burden Values") +
  scale_fill_viridis_b()

b = ggplot(data = eji_map, aes(fill = rpl_mod2)) +
  geom_sf(lwd = 0.1) +
  theme_void() +
  labs(title = "Social Vulnerability", fill = "Social Vulnerability Values") +
  scale_fill_viridis_b()

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c = ggplot(data = eji_map, aes(fill = rpl_mod3)) +
  geom_sf(lwd = 0.1) +
  theme_void() +
  labs(title = "Health Vulnerability", fill = "Health Vulnerability Values") +
  scale_fill_viridis_b()
# join maps on one shared document
plt_eji_themes = ggarrange(
  a + theme(legend.position = "none"),
  b + theme(legend.position = "none"),
  c + theme(legend.position = "none"),
  ncol = 3)
annotate_figure(plt_eji_themes, top = text_grob("Map 6: Maps of Environmental Justice Index T

```